Understanding Corporate Coherence Theory and Evidence

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Abstract

Multiproduct firms are perceived to be coherent in their scope, yet there is no strong theoretical foundations to explain coherence in modern industrial organization theory. This paper shows that as U.S. manufacturing firms grow more diverse, they maintain a constant level of coherence between neighboring activities. This finding runs counter to the idea that firms with many activities are generally more 'incoherent'. A framework is then presented which appeals to the nature of enterprise learning, path dependencies, and the nature of the selection environment to explain the ubiquity of coherent diversifiers.

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Introduction

There are three relatively unexplored characteristics of the boundaries of the modern corporation that deserve an explanation: (i) their multiproduct scope, (ii) the non-random (coherent) distribution of product portfolios inside firms, and (iii) the relative stability in the composition of firms' product portfolio over the long run. [1] In short, it appears that there is some coherence in the ways firms diversify, and this coherence is relatively stable over time. Thus Shell Oil Company has been in the oil business exploration, production, refining, and marketing of crude oil and petroleum products - for almost a century. It has diversified into petrochemicals and little else. IBM has been in information technologies for almost half a century; Intel is focussed in semiconductors; ICI and DuPont are primarily in chemicals; and Boeing is in aerospace. Put differently, it is intriguing, from a theoretical perspective if not from a practical one, that Boeing does not make buses, DuPont does not make airplanes, ICI does not own supermarkets, IBM does not make cars, and Shell Oil does not make jewelry.

The standard neoclassical theory of industrial organization [e.g., Tirole (1988)] is almost completely silent with respect to the matter of corporate coherence, despite its obvious salience in industrial organization. In this paper we attempt to measure the coherence of business firms, and

then outline a theory of corporate coherence. In order to do so, we pay explicit attention to enterprise learning and the development of competences, but also to complementary assets, technological opportunities and the strength of competition in product and capital markets. These concepts are alien to much of the received industrial organization literature, but we suggest that they have utility in unlocking a significant conundrum in modern industrial organization theory.

The Concept of Coherence

Firms are coherent to the extent that their constituent businesses are related to one another. In the language of economics, businesses are related if there are economies to their joint operation and/or ownership. In this paper we focus not on why firms diversify, [2] but on why they diversify coherently. [3] Setting aside for the moment the conglomerate form of business organization, what is remarkable about firms, as we will show, is the relative 'coherence' in their business activities. [4] Admittedly, the boundaries of the firms are sometimes quite fuzzy, with interfirm agreements, joint ventures, and consortia rendering precise delineations of a firms' activities imprecise.

Few if any business firms are literally single product. In many cases, however, diversification is a straightforward consequence of micro-level diversity in the demands of the buyer population, such as that which exists for different ice cream flavors, shoe sizes, pizza toppings, and airline destinations. Much of this diversity is so readily accommodated on the production side that we find it useful to view such firms as being single product. It is the more variegated product diversity that we wish to explain, such as when a manufacturer of microprocessors also makes memory devices.

A salient attribute of diversification is that, as we shall show, firms over time add activities that relate to some aspect of existing activities. They build laterally on what they have got. New product lines bear certain technological and market similarities with the old. Thus the sequence is generally for firms to begin as single product and subsequently become multiproduct, rather than the other way around; and new product lines very often, though certainly not always, utilize capabilities common with existing product lines. Indeed, many prescriptions with respect to diversification strategies are characterized by their emphasis on the desirability for firms to 'stick to their knitting, [5] though such prescriptions rarely specify what the knitting is to which firms should stick.

Corporate diversification is also a phenomenon that can expand and contract. Firms not only add businesses, they also commonly divest. In the United States, the 1980s witnessed firms actively engaged in divestitures, including ITT, Allegis, and RJR Nabisco. Indeed, there often appears to be a degree of circularity to the fashion in which new businesses are added and subsequently divested. This is particularly so when corporations diversify through acquisition. [6] On the other hand, corporations often display remarkable similarities with respect to their diversification strategies. Almost everywhere the auto companies have diversified into a wide range of vehicles, e.g., Mercedes, Fiat, GM, and Renault all make cars, trucks, buses, and tanks. And, with few exceptions, aerospace companies have remained relati- vely focussed. The challenge is to develop a theory to explain

1Pavitt, Robson and Townsend (1989, table 5, p. 91) show that the sectoral priorities of U.K. firms have been relatively stable over a 40 -year period. Chandler's (1990) appendices suggest the same is true for U.S. and German firms. [2] Teece (1980, 1982) has articulated a plausible theory of the multiproduct firm based on transaction costs, excess resources, and scope economies. But in these earlier papers he is not concerned explicitly with the coherence issue. [3] In this paper we are not explicitly interested in vertical integration questions, which have been handled elsewhere, most notable in Williamson (1985).

[4] We recognize that in the popular press the scope of diversification is frequently remarked upon; but unless one has a preconceived view of what the corporation ought to look like, one should find coherence equally remarkable. Certainly in the standard textbook on microeconomic analysis of firms there is usually nothing to explain why

not only coherence, but also similarities and differences in the nature of coherence across firms and industries.

In this regard, it is useful to differentiate between coherence at the business level and coherence at the corporate level. There is obviously much greater coherence and stability at the former level. Since its formation in 1918, Hertz has had at least five different corporate parents, yet its car rental business has been basically impervious to the identity of its 'parents'. In this paper we are interested in both business unit coherence and corporate coherence. The latter will necessarily take us into a consideration of various corporate forms, including the conglomerate.

Coherence, we should point out, is different from 'specialization'. Specialization refers to the performance of particular tasks in a particular setting. This is the manner in which Adam Smith implicitly defined the concept, and it is the way we will use it. Thus in the famous pin making example, Smith referred to specialization in terms of particular tasks, like grinding the point. Thus a firm may exhibit coherence though it may not necessarily be specialized. Specialization is a special case of coherence when the coherence is confined to a single product line. We are, accordingly, defining coherence in a multi-product sense.

A firm exhibits coherence when its lines of business are related, in the sense that there are certain technological and market characteristics common to each. A firm's coherence increases as the number of common technological and market characteristics found in each product line increases. Coherence is thus a measure of relatedness. A corporation fails to exhibit coherence when common characteristics are allocated randomly across a firm's lines of business.

Measuring the Coherence of Business Firms

3.1. Prior literature

A great deal of work has been carried out on the measurement of corporate diversification, [7] but much less has been directed at the measurement of inter-business relatedness. One approach, developed by Caves (1981), is to use the hierarchy implicit in the SIC system, so that businesses in different 4 -digit industries but the samc 3 -digit industry are 1 'unit' apart, whereas businesses whose closest connection is their 2 -digit industry memberships are 2 'units' apart, and so forth. Lemelin (1982) measured inter-industry related- ness as the correlation coefficient across input structures taken from the input-output table. Klavans' (1989) index of technology relatedness is also based on input patterns, but he uses the amount of overlap in occupational categories as an index. A fairly noisy measure of relatedness has been developed by Gollop and Monahan (1989): they compute the dissimilarity between the patterns of input shares, using nine classes of input (i.e., production wages, fuel, purchased services, etc.). Working with Census of Manufacturers data, Gollop and Monahan argue that because the data are 'filtered through the SIC system, the underlying notion of product dissimilarity must have a supply-side orientation ... differences in distribution systems and/or marketing patterns are ignored. It follows that the formulation of product heterogeneity must be based wholly on technical considerations' (1989:4).

One of the contributions of the present study is to develop an index using SIC data that goes

firms are diversified, let alone the nature of their diversification. [5] See Peters and Waterman (1982). [6] Thus the conglomerate ITT was put together by Harold Geneen, only to be dismantled by his successors. Similarly, Richard Ferris put together the Allegis Group, starting with United Airlines, only to have it undone months later by his successors.

[7] Gort (1962) used specialization measures, Rumelt (1972) combined specialization with adjudged relatedness, Berry (1971) developed the Herfindahl measure, which Palepu (1975) identifies as entropy.

beyond input considerations and overturns Gollop and Monahan's dictum. This is accomplished by moving away from associating the properties of SIC industries, and instead looking directly at the frequencies with which they are combined in firms.

3.2. A survivor measure of relatedness

The survivor principle is that economic competition will lead the disappearance of relatively inefficient organizational forms. Stigler (1968) argued in favor of using this principle to measure economies of scale (or estimate optimum firm size), noting that 'not only is the survivor technique more direct and simpler than the alternative techniques for the determination of the optimum size of the firm, it is also more authoritative' (1968:74). To apply the spirit of this principle to the study of relatedness, we assume that activities which are more related will be more frequently combined within the same corporation. Thus, if corporations which engage in activity A almost always also engage in activity B, we would conclude that these activities are highly related. Conversely, activities that are rarely or never combined are unrelated.

To be explicit, let the universe being studied consist of K corporations each active in two or more of I industries. Because undiversified companies shed no direct light on the relatedness of industries, they are omitted from consideration. To describe this data let $C_{ik} = 1$ if corporation k is active in industry i, and 0 otherwise. We call the operations of a corporation in an industry an activity. Thus, both corporations and industries are comprised of activities. The number of activities in corporation k is $m_k = \sum_i C_{ik}$ and the number of diversified activities in industry i is $n_i = \sum_k C_{ik}$.

Now consider the number J_{ij} of corporations which are active in both industry i and industry j, where

$$J_{ij} = \sum_k C_{ik} C_{jk}$$

This count of joint occurrences of industries i and j can be used to construct a measure of relatedness. One initially appealing choice is $P(j/i)/J_{ij}/n_i$, the conditional probability that a corporation is active in industry j given that it is active in industry i. Unfortunately, $P(j/i) = P(i/j)$ because (in general) $n_i = n_j$.

The approach taken here is to compare the observed value of J_{ij} to the value that would be expected under the hypothesis that diversification is random. The intuition is straightforward: J_{ij} increases with the relatedness of i and j, but it also can be expected to increase with n_i and with n_j. Thus, if n_i and n_j are large, one would expect to see fair number of corporations active in both industries even if there was little relatedness between them. Conversely, if n_i or n_j is small, one would not expect to see very many linkages even if there were substantial synergy. Hence, the information in J_{ij} about relatedness can be extracted by comparing it to the number of linkages that would be observed for a given n_i, n_j and K were there no relatedness, i.e., if industries were randomly assigned to companies.

To operationalize the random hypothesis, assume that n_i and n_j are fixed. A sample (without replacement) of size n_i is drawn from the population of K corporations and those chosen are assigned activities in industry i. A second independent sample (without replacement) of size n_j is then drawn from the population of K corporations. Those chosen are assigned activities in industry j. Given this scheme, the number x_{ij} of corporations active in both industry i and j is a hypergeometric random variable (population K, special members n_i, and sample size n_j). That is,

$$\Pr\left[X_{ij} = x\right] = f_{hg}\left(x, N, n_i, n_j\right) = \frac{\binom{n_i}{x}\binom{K-n_i}{n_j-x}}{\binom{K}{n_i}}$$

The mean and variance of X_{ij} are

$$\mu_{ij} = E\left(X_{ij}\right) = \frac{n_i n_j}{K} \quad \sigma_{ij}^2 = \mu_{ij}\left(1 - \frac{n_i}{K}\right)\left(\frac{K}{K-1}\right)$$

If the actual number J_{ij} of linkages observed between industry i and industry j greatly exceeds the expected number, μ_{ij}, the two industries are highly 'related'. Accordingly, the measure of relatedness is taken to be

$$t_{ij} = \frac{J_{ij} - \mu_{ij}}{\sigma_{ij}}$$

In analogy to a t-statistic, i_{ij} measures the degree to which the observed linkage between the two industries exceeds that which would be expected were the assignments of industries to companies simply random.

3.3. The data

The data used in this study were obtained from the 1987 Trinet Large Establishment tape. On this tape Trinet attempts to provide information on every establishment in the United States with over 20 employees. Each of the approximately one-half million records on the tape describes an establishment: its name, address, telephone number, employee count, its primary fourdigit SIC (1987 standard), estimated value of shipments, and corporate parent, if any. After aggregating a parent's establishments in the same industry, deleting administrative units, deleting government and non-profit 'industries', and deleting corporations that were not diversified, 18,620 companies remained. These diversified firms were active in 958 different 4digit SIC industries, with a total of 66,688 activities represented.

Of the 448,403 possible linkages between pairs of industries, 128,338 were observed. A t_{ij} measure of relatedness was calculated for each such pair. It ranged from -4.02 to 111.3 . The average relatedness was 4.33 and the standard deviation was 5.41 .

The highest observed relatedness was between SIC 5181 and SIC 5182; between beer & ale wholesalers and wine & spirits wholesalers. In this case, there were 307 SIC5181 activities and 316 SIC5182 activities in multiactivity firms; 255 of these appearances were joint. Some other examples of high relatedness illustrate how the measure overcomes the vagaries of the SIC system:

1. The 9 cane sugar refining (SIC 2061) activities and 7 beet sugar refining (SIC 2062) activities appeared jointly 7 times, inducing a relatedness of 85.9.

2. The 33 ordnance & accessories (SIC 3489) activities and the 16 military tanks (SIC 3795) activities appeared jointly 10 times, inducing a relatedness of 59.3.

3. The 81 fabric gloves manufacturing (SIC 2381) activities and the 13 leather glove manufacturing activities (SIC 3151) appeared jointly 6 times, inducing a relatedness of 50.7.

4. The 81 natural gas liquids (SIC 1321) activities and the 117 petroleum refining (SIC 2911) activities appeared jointly 39 times, inducing a relatedness of 54.2.

3.4. Measures of the coherence of firms

A diversified (multi-activity) corporation consists of m activities. [8] Activity i has e_i employees and relatedness [9] τ_{ij} with activity j. The weighted-average relatedness WAR_i of activity i to all other activities within the firm is defined as

$$WAR_i = \frac{\sum_{j \neq i} \tau_{ij} e_j}{\sum_{j \neq i} e_j}.$$

This measure indicates the degree to which activity i is related to all of the other activities of the firm, where relatedness is weighted by employee count. Another interpretation is that WAR_i is as the expected relatedness between activity i and the activity housing an employee randomly selected from elsewhere in the firm.

The second measure of intra-firm relatedness captures the strength of association between activity i and its closest neighbors. In a corporation with m activities there are $m(m-1)/2$ different pairs of relatedness measures among activities. However, only $m-1$ links among activities need be drawn to create a connected graph that includes them all. The graph with $m-1$ links such that the sum of the relatedness measures on each link is largest is called the maximum spanning tree. Such a tree is defined by $\lambda_{ij} = 1$ if a link between activity i and activity j is part of the tree, and 0 otherwise. Given the maximum spanning tree of a company's activities, the weighted average relatedness of neighbors to activity i is defined by

$$WARN_i = \frac{\sum_{j \neq i} \tau_{ij} e_j \lambda_{ij}}{\sum_{j \neq i} e_j \lambda_{ij}}.$$

3.5. Findings

These measures were computed for each of the 41,857 activities in a Trinet multi-activity manufacturing company. [10] Table 1 shows average values of WAR_i and $WARN_i$, as well as standard deviations, for various values of m.

As expected, the value of WAR_i falls with m : the regression [11] of m on WAR_i yields $R^2 = 0.0118$ and a slope coefficient of 0.091 .

These results suggest that as U.S. manufacturing firms grow more diverse, they maintain a constant level of (local) coherence between neighboring activities. This finding runs counter to the idea firms with many activities are necessarily unrelated or incoherent. Taking the dyadic firm (a firm active in only two activities) as the standard, the 6,972 dyadic manufacturing firms have an average inter-activity relatedness of 19.8 (table 1). In firms with 3 activities, the average relatedness among neighbors increases slightly to 21.5 . 10 activities increases average relatedness among neighbors to 23.4 and with 70 or more activities it increases to 27.5

These data reveal an industrial fine structure that does not vary a great deal (on average) with the diversity of the corporation. A useful analogy would be to a residential neighborhood, where $WARN_i$ measures the inverse of the average distance from a house to its neighbors and WAR_i measures the inverse of the average distance from a house to all other houses in the area. As

[8] The k th corporation has m_k activities; the subscript is suppressed here to focus on the relationships among activities within a single firm. [9] Note that $\tau_{ij} = t_{q(i)q(j)}$, where the corporation's i th activity is in industry $q(i)$. [10] Characterizing companies by the 2 -digit SIC receiving the most employment, manufacturing companies have 2 -digit SIC codes in the range 20-39.

[11] All regression results reported are significant at the 0.0001 level. The high level of significance reflects the impact of 41,857 observations rather than large effects.

Table 1 Relatedness of activities, establishments with over 20 employees.

		Coherence measures			
		WARN $_i$		WAR_i	
Number of activities (m), 1987	Obs.	Avg.	Std. Dev.	Avg.	Std. Dev.
2	13,944	19.8	17.6	19.8	17.6
3	4,434	21.5	17.6	18.1	15.4
4	2,748	24.7	18.8	19.0	15.5
5	2,010	24.1	17.5	17.1	13.5
6	1,386	23.4	16.7	15.5	12.2
7	1,183	24.7	16.4	15.5	11.4
8	1,016	23.6	15.4	14.6	10.9
9	801	23.5	15.5	13.4	10.2
10	760	23.4	14.6	13.0	8.9
11	594	24.0	15.3	12.8	9.3
12	588	23.6	15.2	11.7	8.0
13	455	24.9	16.2	11.3	8.5
14	616	23.7	13.5	12.0	8.4
15	345	23.9	15.2	12.1	8.7
16	448	22.2	13.3	10.5	8.1
17	306	20.6	11.0	10.0	6.4
18	378	23.4	13.5	11.3	8.1
19	342	25.8	13.5	11.7	7.6
20-29	2,975	25.0	13.9	10.2	7.0
30-39	2,032	24.0	12.8	9.0	5.6
40-49	1,598	25.4	13.3	8.5	5.4
50-59	921	28.3	13.8	8.9	5.4
60-69	773	28.2	14.7	9.1	5.7
≥ 70	1,204	26.8	14.2	6.8	3.8

the area covered is increased, the distance to nearest neighbors does not change, but the average distance to all other homes does rise. In the industrial landscape, one sees that in a more diverse firm, just as in a larger neighborhood, the average distance to other activities increases, but the average distance to neighbors does not.

Our findings suggest that if firms grow more diverse, they add activities that relate to some portion of existing activities. Furthermore, the data suggest that the strength of relatedness between new and old business does not change much as firms grow more diverse. Coherence is indeed a salient characteristic of the American business enterprise. However, there is variability in the data, and much of the rest of this paper will attempt to build a theory to explain differing levels of coherence across firms by appealing to aspects of a firm's learning, history, and selection environment.

Elements of a Theory of Corporate Coherence

4.1. The limitations of existing theories of the firm

Neoclassical theories of industrial organization, including the so-called new industrial organization [e.g., Tirole (1988)], have little to say about coherence and do not readily explain the findings identified in section 3. They neither explain coherence nor its absence. While there is some development of market power and economies of scope explanations for diversification, these theories are without strong organizational implications. Thus one can 'explain' diversification through notions of 'full line forcing' or 'economies of scope', but such theories rarely explain why a set of contractual relationships among specialist firms could not accomplish the same objectives. For instance, consider economies of scope. If the production of x_1 and x_2 involves scope economies, then $c(x_1, x_2) < c_1(x_1, 0) + c_2(0, x_2)$. However, in standard treatments the organizational dimensions of the joint cost function $c(x_1, x_2)$ are never explained. The amalgamation implied by the cost function could be accomplished by a variety of organizational mechanisms, including contracts [Teece (1980, 1982)]. A theory of multiproduct organization cannot be built on the concept of economies of scope. Nor can a theory of corporate coherence.

Other approaches are almost equally as barren. For instance, ecological theories of economic organization provide little mileage, though they may provide a starting point. Ecological approaches are sometimes quite explicit about the limited ability of firms to diversity. Thus, Freeman and Boeker (1984) remark, '[I]ndividual firms are not often entirely free to generate unique strategies. There is a strategic choice involved when one decides to open one kind of business or another. This choice is more or less fixed at firm founding and is constrained by an environment which permits less than free will in strategic choice. A bank selling hamburgers at some of its teller windows would be viewed by all as inconsistent' (p. 71). Likewise, transaction cost economics does not provide much of a foundation for a theory of diversification. Rather, it is from the organizational economics [12] perspective that we draw our theory of corporate coherence. Fig. 1 maps various theories of the firm and identifies the intellectual thrusts associated with different approaches.

We advance the fundamental proposition that the boundaries of the corporation can be understood in terms of learning, path dependencies, technological opportunities, the selection environment, and the firm's position in complementary assets. (Figs. 2a and 2b indicate the degree of coherence we expect to see in different business environments with different technological trajectories). To come to grips with coherence issues, we find it necessary to put to the side theories of the firm which consider it as a nexus of contracts involving labor, management, and suppliers; we also find theories which model the firm as a production function quite unhelpful. The former characterization puts zero weight on the organizational and administrative capabilities of firms; the latter assumes the existence of organizational and administrative capabilities which in our view need to be explained. This is because while the production function purports to be a technological relationship between inputs and outputs, unless the production relationship so described is contained wholly within a machine, issues of organizational and individual skills will necessarily be involved. Each of the key elements of a theory of corporate coherence will now be explained.

4.2. Enterprise learning [13]

A fundamental characteristic of economic activity is that it provides the opportunity for learning. Learning is a process involving repetition and experimentation which enables tasks to be performed better and quicker, and new production opportunities to be identified. As Simon (1991) notes, 'All learning takes place inside individual human heads; an organization learns in only two ways: (a) by

the learning of its members, or (b) by ingesting new members who have knowledge the organization didn't previously have. But what is stored in any one head in an organization may not be unrelated to what is stored in other heads; and the relation between those two (and other) stores may have a great bearing on how the organization operates' (p. 125). Enterprise learning has several key characteristics.

First, it is generally cumulative. What is learned in one period builds upon

Fig. 1. Authors and theories of the firm.

School	Representative Authors	Intellectual Thrust	Coherence Addressed
	Fama (1980); Fama & Jensen (1983)	Principal agent/contract nexus	No
	Grossman & Hart (1982)	Residual claimant	No
Neoclassical Economics	Milgrom & Roberts (1987)	Influence costs	No
	Baumol, Panzar & Willig (1982)	Subadditive cost function	No
	Alchian & Demsetz (1972)	Information, monitoring	No
Ecological Theory	Hannan & Freeman (1977)	Environmental selection	No
Evolutionary Theory	Nelson & Winter (1982)	Routines/selection	No
Transaction Cost Economics	Coase (1937)	Transaction and communication cost	No
	Williamson (1975, 1985)	Asset specificity, incentive limits	No
Organizational Economics	Chandler (1977, 1990)	Historical process & path dependencies	Yes
	Teece (1980, 1982, 1986)	Technology transfer costs, learning, specific assets, technological opportunities, appropriability	Yes

Fig. 2a. Corporate coherence matrix (tight selection environment).

		Tight Selection	
		Slow Learning/ Restricted Opportunities	Rapid Learning/ Rich Opportunities
Evolutionary Path**	Wide	Coherent Diversifiers (low growth)	Coherent Diversifiers* (high growth)
		Single Product and Vertical Integration (low growth)	Single Product (high growth)
	Converging	Lateral and Vertical Network Firms (low growth)	Lateral and Vertical Network Firms (high growth)

Coherent diversifiers might be thought of as laterally integrated firms. ** Note: Evolutionary paths are considered to be shaped by a multitude of factors discussed in section IV.B, including the firm's core competences, its complementary assets, and the technological opportunities lying ahead. Because learning is seen as local, all paths are seen at a point in time as being narrow; over time, firms may be able to diversify quite broadly, all the while preserving the local coherence identified in III.E.

The organizational forms depicted here can co-exist with the more robust forms in Figure 2a because the selection environment is assumed to be weak. what was learned in an earlier period. Individual and group knowledge gaincd through lcarning must, however, be constantly used in order to be preserved. There may be switching costs associated with moving among tasks due to the set-up time associated with moving to a new task and 'forgetting' which occurs before returning to the old.

[12] See Barney and Ouchi (1986), chapter 1. [13] A useful compendium of the literature on organizational learning is the special issue of Organizational Sciences, 2:1(1991).

		Weak Selection	
		Slow Learning/ Restricted Opportunities	Rapid Learning/ Rich Opportunities
	Wide	Conglomerate (low growth)	Hollow Corporation (high growth)
Evolutionary Path		Conglomerate (low growth)	Hollow Corporations (high growth)
	Converging	Conglomerate (low growth)	Hollow Corporations (high growth)

Fig. 2b. Corporate coherence matrix (weak selection environment) (additive with fig. 2a*).

Business learning involves organizational rather than individual skills. [14] While individual skills are of relevance, their value depends upon their employment in particular organizational settings. Learning processes are intrinsically social and collective phenomena. Learning occurs not only through the imitation and emulation of individuals as with teacher-student or master-apprentice, but also because of joint contributions to the understanding of complex problems. Learning requires common codes of communication and coordinated search procedures.

The knowledge generated by such activity may be thought of as residing in organizational 'routines' [Nelson and Winter (1982)] and standard operating procedures. These are patterns of interactions which represent successful solutions to particular problems and which are resident in group behavior, though certain subroutines may be resident in individual behavior. Because of the complexity of such behavior, knowledge embedded in routines cannot be fully captured in codified form. That is, it has a tacit dimension that often cannot be readily articulated. Hence, it is the routines themselves, and the ability of management to call upon the organization to perform them, that represents an enterprise's essential capability.

Routines can be of several kinds. Static routines embody the capacity to replicate certain previously performed tasks. Needless to say, such routines are never entirely static, because with repetition routines can be constantly improved. The presence of standard learning curve economies often indicate the operation of static routines. Dynamic routines are directed at learning, and new product-process development. Thus R&D activity proceeds through the employment of routines to ascertain where to probe, how to probe, and how much to probe.

Because routines involve a strong tacit dimension, they may not be easy to imitate. To the extent that this is so, routines contribute to a firm's distinctive competences and capabilities. Such capabilities, by virtue of their evolution in particular environment and organizational contexts, are likely to differentiate firms from each other and provide the basis for differential performance vis-a-vis competitors.

As discussed below, the rate and direction of enterprise learning is partly a function of the tcchnological opportunities which are available to a firm. These may be rich because of recent breakthroughs in basic science. Upstream innovation by supplier firms might also enrich the environment for firms downstream, and vice versa, as is true in the personal computer industry. Here innovation in microprocessors and in memory devices con- stantly expands the price performance of products which embody them. Enterprise learning may have a strong 'public' element - as is the case when independent suppliers are responsible for innovation - thereby supplementing proprietary learning. When learning has a strong proprietary element, individual firms can significantly pull away from the public knowledge base in terms of their own capabilities.

Needless to say, learning is rarely entirely exogenous. While the external opportunity set may be driven primarily by public investments in basic science, individual firms can themselves push their

[14] Simon (1991) also notes that 'human learning in the context of an organization is very much influenced by the organization, has consequences for the organization, and produces phenomena at the organizational level that go beyond anything we would simply infer by observing learning processes in isolated individuals' (p. 126).

own and the industries' opportunity through investment in early stage (basic and applied) research. Furthermore, firms in the same industry may have rather different rates of learning because of differences in the human skill base as well as differences in managerial and organizational systems.

A property of organizational learning is that it enables the enterprise as a whole to surmount the bounded rationality of particular individuals. This does not come automatically. Enterprises need to set up mechanisms for the timely capture of performance feedback so that success and failures become identified and provide the foundation of learning experiences. Furthermore, a corporate culture which identifies, supports, and rewards learning is almost a prerequisite for rapid learning to occur.

4.3. Path dependencies

Local Learning and the Textbook Paradigm. The local nature of enterprise learning significantly restricts what firms can do. Their future activities are highly dependent on what they have done in the past. Yet in economics textbooks firms have an infinite range of technologies from which they can choose, and markets they can occupy. Changes in product or factor prices will be responded to instantaneously, with technologies moving in and out according to value maximization criterion. Only in the short run are irreversibilities recognized. Fixed costs - such as equipment and overhead cause firms to price below fully amortized costs but never constrain future investment choices. 'Bygones are bygones' in this conceptualization, and firms are able to strike out in directions which market signals suggest are attractive, no matter how distant those directions from a firm's current line of business.

Oliver Williamson (1975, 1985) has done much to enrich the standard treatment by introducing transaction-specific assets. Such assets have idiosyncratic uses and cannot be redeployed without substantial loss of value. Williamson demonstrates how the need for such assets generates particular requirements with respect to an industry's governance apparatus. Generally vertical integration or some other form of organizational safeguard is necessary to facilitate such investments and accomplish efficient supply. He is interested in how specific assets impact organizational design; his framework does not directly address how the presence of specific assets molds the market options open to the firm.

The notion of path dependencies goes even further than notions of irreversibilities employed in transaction cost economics. It recognizes that 'history matters'. Thus a firm's previous investments and its repertoire of routines (its 'history') constrains its future behavior. This follows because learning tends to be local. That is, opportunities for successful new developments will be 'close in' to previous activities and will thus be transaction and product specific [Teece (1988)]. This is because learning is a process of trial, feedback, and evaluation. If too many parameters are changed simultaneously, the ability of firms to conduct meaningful natural quasi experiments is attenuated. If many aspects of a firm's learning environment change simultaneously, the ability to ascertain cause-effect relationships is confounded because cognitive structures will not be formed and rates of learning diminish as a result.

Two key aspects of the learning environment for new product development are the technologies being employed, and the market into which the new product is to be launched. If firms attempt to enter new markets with new technologies, failure is likely to be the norm because the effort is likely to be outside the firm's learning range. Fig. 3 attempts to characterize this diagrammatically. This possibly explains why firms which are defense contractors have problems competing in civilian markets even if they are technologically sophisticated. Northrup provides a good example. Either technological or market distance can keep a firm penned up.

Moreover, a firm stationary for some period in the rapid learning range will of course be able to subsequently migrate (incrementally) to products and activities at greater technological and market distance. Where it can go will, however, be a function of where it has been. This is contrary to conventional economics treatments. In textbook treatments one is in theory able to forecast accurately the activity of the firm as a smoothly shifting solution to the analytical equations governing prices and factor costs. History, in the conventional treatments, is not particularly important; the firm is not constrained by where it has been [David (1988)].

What is true at the operational level is also true at the strategic level. The cognitive limits of managers and the costliness of information systems restrict the range of businesses and products that most managers can understand. [15] The limited capability of information systems to render useful information and the limited ability of managers to understand multiple competitive

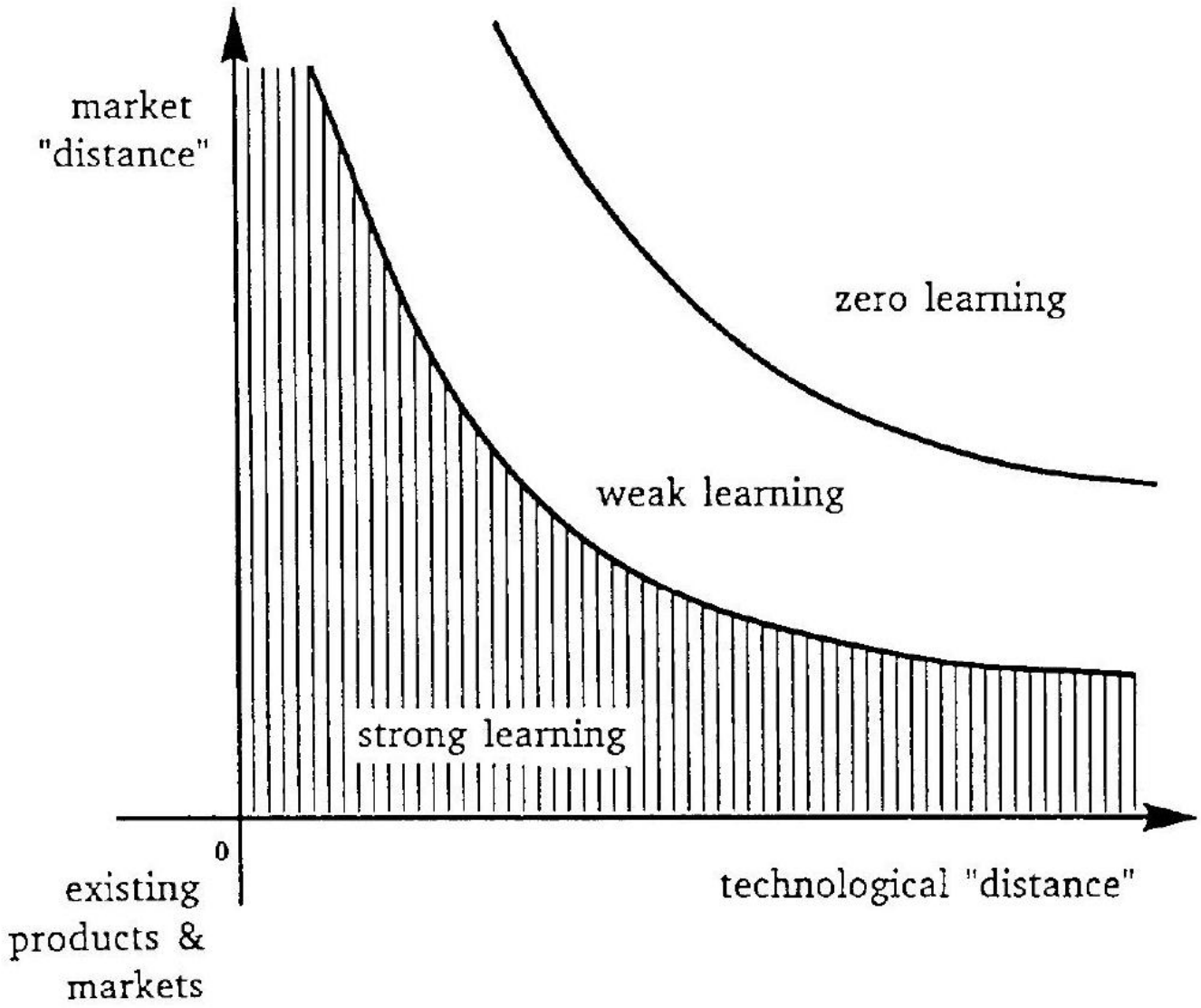

Fig. 3. Firms' (one period) learning domain.

[15] This phenomena perhaps forms the foundation of Quinn's (1980) focus on 'logical incrementalism' in which executives bring about step-by-step movements towards ends which initially are broadly conceived, but which are then constantly refined and reshaped as new information appears.

environments appears to be a binding constraint, particularly in dynamically competitive rnarkets. This is not only because the dynamics of technological change and market evolution are always complex, but also because the dynamics are likely to vary markedly across industries and product lines and because long lags may be necessary to capture the performance feedback necessary for learning.

The Concept of Competence. A firm's competence [Teece (1988)] is a set of differentiated technological skills, complementary assets, and organizational routines and capacities that provide the basis for a firm's competitive capacities in one or more businesses. 'These people are good at X ' summarizes external perceptions as to the nature of these competences. In essence, competence is a measure of a firm's ability to both solve technical and organizational problems. It isn't product specific. The presence of such competences open possibilities for multiproduct activity. [16]

A core competence will typically have both an organizational/economic and a technical dimension. Unfortunately, economic analysis typically separates the two. Thus Frank Knight (1921) insisted that economic problems be split from technical ones, with only the former being the proper domain for economic analysis. The profession appears to have agreed. However, since the two are initially interwoven in practice, splitting the two will stunt if not cripple efforts to theorize meaningfully about the firm and economic organization more generally. [17]

Organizational/economic competence involves: (1) allocative competence deciding what to produce and how to price it; (2) transactional competencedeciding whether to make or buy, and whether to do so alone or in partnership; and (3) administrative competence - how to design organizational structures and policies to enable efficient performance. Technical competence, on the other hand, includes the ability to develop and design new products and processes, and to operate facilities effectively. It also involves the ability to learn. Typically, such competences have an important tacit dimension, making replication by others difficult but not impossible. Indeed, when a core competence exists, its replication proceeds either through deliberate efforts directed at internal expansion or through the exit of key individuals who collectively take certain competences with them, and then raise capital to buy the necessary complementary assets.

In our view, the competitive strength of a particular corporation is a function of its underlying technical and organizational competences. [18] The existence of organizational competences explains why plant and equipment produces more when owned by one company rather than another. The value of Tobin's Q - the difference between the market value of a firm's securities and the replacement cost of its assets - may reflect the presence of either technical or organizational competences. Technical competences may be more product specific than organizational ones.

Great companies like Hewlett Packard, DuPont, Exxon and Toyota consist of clusters of such technical-organizational competences. Successful companies must possess both organizational and technological competences. As described below, hollow corporations lack technical competences;

[16] Whether a firm has a sufficient foundation for efficient multiproduct activity will depend on the ease with which the competence in question can in fact be transferred across product lines [Teece (1980)], the complementary assets the firm possesses, and the opportunity cost of the activity. The latter will depend in part on the technological and market opportunities lying ahead in the firm's traditional lines of business.

[17] Economic theory almost always assumed economic competence. It is embedded in the implicit (and sometimes explicit) assumption that cconomic agents are generally omniscient, or at least hyper-rational. To admit that organizational/economic competence may be scarce undermines a very large body of neoclassical economic theory. For an excellent discussion of this and economic competence more generally, see Pelikan (1988). Pelikan's treatment has helped shape our views on economic competence. [18] Numerous recent studies compare the operating performance across companies with respect to quality, speed, capital utilization and other measures of efficiency. Examples include Garvin's (1988) study of room air conditioners that showed dramatic differences among firms in defect rates. Such differences appear to be the result of different systems and procedures. Clark and Fujimoto (1991) in studying the auto industry found high variation in the efficiency of product development process and the quality of the result-

business is performed through contractual agreements struck with others. When the contracts terminate, the firm's ability to deliver value evaporates. We define a hollow corporation as a business entity that does not have any core technical competences and uses contractual mechanisms to link particular market requirements with productive capacities. Thus a hollow corporation may manage a nexus of contracts for design, manufacture, distribution, and service. Such enterprises often turn out to have weak survival properties because the basis of their initial advantage is easy-to-imitate organizational competences.

Likewise, a conglomerate is a portfolio of autonomous business units held together in a holding company. It is a 'mutual fund' of businesses without organizational competences at the corporate (holding company) level. [19] As discussed later, we do not believe that organizations such as these have survival properties unless they are somehow shielded from product and capital market competition. They are likely to survive longer than hollow corporations, not because they are more efficient, but because they are likely to be able to utilize cash flows generated by other divisions; hollow corporations more quickly become unglued if a subunit fails to perform.

Complementary Assets. The existence of complementary assets [Teece (1986)] helps contour evolutionary paths, and evolutionary paths help determine the composition of a firm's portfolio of complementary assets. Prior commercialization activities require and enable firms to build complementary assets. Such assets, built to support the firm's prior activities, may have other uses as well. Rarely are such assets completely specialized to a particular product. Examples include distribution systems, manufacturing plant and equipment, and complementary technologies. Complementary assets typically lie downstream from product-process development in the value-added chain.

New products and processes can either enhance or destroy the value of complementary assets [Tushman and Anderson (1986)]. Thus the development of computers enhanced the value of IBM's direct sales force developed for other office products, while disc brakes destroyed the value of the auto industries' investment in drum brake facilitics.

The presence of complementary assets, particularly those that can be enhanced by innovation, helps steer the evolution of new technologies [Teece (1986), Mitchell (1989)]. The reason why Singer Sewing machine got into the furniture business was in large measure due to the fact that it had designed and manufactured sewing machine cabinets to help facilitate the sale of sewing machines. Thus, a particular firm's path dependencies are not just technological determined. Paths are identified by technological paradigms and trajectories [Dosi (1982)] and constrained further by the complementary assets the firm has already built or can readily acquire.

In the world of neoclassical microeconomic theory, sunk costs, whether core or complementary, often do not matter. We suggest that in reality they do because they suggest a strategy for rapidly improving earnings - namely to increase the utilization rate. This steers firms with excess capacity in their complementary assets toward searching for value-creating opportunities to take up the slack [Teece (1980, 1982)]. Put differently, the existence of slack is likely to trigger search processes directed at raising the utilization rates of existing assets, simply because the cognitive path associated with doing this is likely to be clearer than other alternatives.

Technological Opportunities. The concept of path dependencies can be given forward meaning through the consideration of an industry's technological opportunities. It is well recognized that how far and how fast a particular area of industrial activity can proceed is in part due to the technological opportunities that lie before it. Such opportunities are usually a lagged function of

ing product. Jakumar (1988) showed significant differences in performance between flexible manufacturing systems (FMS) in Japanese and American firms.

[19] By definition, a conglomerate has a limited headquarters function and so cannot have significant organizational competence at the corporate level.

foment and diversity in basic science, and the rapidity with which new scientific breakthroughs are being made. They may also lie with customer-user inventiveness.

However, technological opportunities may not be completely exogenous to industry, not only because some firms have the capacity to engage in or at least support basic research, but also because technological opportunities are often fed by innovative activity itself. Moreover, the identification of such opportunities is affected by the organizational structures that link the institutions engaging in basic research (primarily the university) to the business enterprise. Hence, the existence of technological opportunities is a necessary but not a sufficient condition for innovation.

More important for our purposes is the rate and direction in which relevant scientific frontiers are being rolled back. Firms engaging in R&D may find the path dead ahead closed off, though breakthroughs in related areas may be sufficiently close to be attractive. Likewise, if the path dead ahead is extremely attractive, there may be no incentive for firms to shift the allocation of resources away from traditional pursuits. The depth and width of technological opportunities in the neighborhood of a firm's prior research activities thus are likely to impact a firm's options with respect to both the amount and level of R & D activity that it can justify.

Convergence of Paths. Technological paths are shaped not only by a firm's history, but also by changes in the public knowledge base. These changes, often driven by developments in basic science, may be such that the technological foundations of businesses shift dramatically. This may cause the technological knowledge base of industries to converge or diverge. Thus digital electronics is presently causing a convergence in the public knowledge base supporting computer and telecommunications firms. This has implications for corporate coherence because products which may be aimed at rather different customer groups might suddenly develop common technological and production roots.

4.4. Selection

Selection is the process by which inefficient firms or businesses get weeded out. At the firm level, selection is accomplished primarily through market processes involving entry and exit; growth and decline provide a second mechanism. Neither mechanism works instantaneously so that firms with different capabilities may exist side-by-side attempting to serve the same customer needs. Indeed, bankruptcy laws in certain countries may slow selection processes. [20] However, 'market rationality' eventually prevails, and less competent firms can be expected to decline and exit. [21]

Thus the survivability of business activity depends on the selection environment and, in particular, the level of competition (both product market and capital market), public policy, and the frequency of technological discontinuities. In weak selection environments, firms and business units are buffered from market rationality. Less efficient firms may be able to survive and possibly even prosper for quite some time. The selection environment for a particular business may be stronger or weaker than that for the corporation as a whole, depending on how corporate management treats internal divisions and products that are not performing up to par.

The availability of free cash flows is perhaps the key regulator of selection inside the firm. If a firm or a business is generating sufficient cash flow to sustain itself and take care of reinvestment needs and opportunities, it will typically be able to survive. If it does not, then it will need additional debt or equity financing. To obtain funding of either kind exposes the firm to capital market discipline that might not otherwise intervene. Hence, initial capitalization, the 'inherited' capital structure of the firm, and subsequent financial restructuring are likely to affect the strength of selection processes with respect to particular business units. We predict the boundaries of the corporation are likely to be drawn 'close in' to core competencies the tighter the selection environment.

Propositions From the Theory

The above framework has powerful implications for business strategy. It suggests that the circumstances under which various organizational forms (implying certain strategic choices) will be viable. It implies that firms can be thought of as integrated clusters of core competences and supporting complementary assets and that the degree of coherence one would expect to observe among the parts at a particular point in time depends on the relationship between learning, path dependencies, opportunities, inherited complementary assets, and selection. We advance the following hypotheses:

Hypothesis One: Specialist Firms

With rapid learning, rich technological opportunities, and tight path dependencies, one will observe 'single' product (specialist) firms growing rapidly. Compaq Computer, Sun Microsystems, Intel, and possibly Boeing and Airbus are good examples. Specialist firms are likely to be young because the probability that technological opportunities in a given area remain rich is likely to decline over time, in part because of new entry.

Hypothesis Two: Vertically Integrated Firms

With slow learning but high path dependencies and specialized assets, one can expect specialist firms with some degree of lateral integration and significant vertical integration emerging over time [22] [e.g. Exxon, Shell, DuPont]. Older firms are likely to be more vertically integrated than young firms because start-ups are less common in industries where learning is slow. [23]

Hypothesis Three: Coherent Diversifiers

With rapid learning, broad path dependencies due to the presence of generic technologies, and tight selection, one can expect to see coherent diversifiers like IBM, BMW, Hewlett Packard, W.R. Gore, and Raychem. Since older firms are likely to have experienced many recessions and periods of retrenchment, they are likely to be coherent diversifiers. Indeed, our findings reported in section 3 suggest this is so. Firms do indeed diversify over time but the modal form across the economy displays diversity but local coherence.

[20] In the early 1990s in the United States, both TWA and Continental Airlines operated for long periods with the protection of Chapter 11 of the Bankruptcy Code. This protection enabled the companies to continue operations despite the fact that they were not able to cover their full costs of operation. [21] There are often exceptions, depending on the selection environment itself.

[22] Note that vertical integration in this framework stems mainly from a growth/'diversification' motive. [23] If a firm is innovating in a regime of weak appropriability, it may find it essential to develop certain complementary assets (vertically integrate) into manufacturing and distribution. This is because with easy imitation of technological competences, marketplace success will swing on the relative positioning of innovator and imitators in the relevant complementary assets and technologies [Teece (1986)]. Thus the best defence of a new product is often a new manufacturing process. The former may be exposed to imitation; the latter may not.

Hypothesis Four: Conglomerates

With low path dependencies, slow learning, and weak selection, we predict the emergence of conglomerates or other highly diversified companies displaying few intracorporate transactions. As selection tightens, such as during recessions, we expect that the most egregious examples of this form will get weeded out. Conglomerates are thus a transitional form.

Hypothesis Five: Network Firms

With rapid learning, colliding technological trajectories and tight selection, one can expect to see incumbent firms becoming enveloped in a dense skein of inter-corporate relationships involving partial equity holdings and joint ventures. [24] Such firms might be called 'network' firms. For example, because of the convergence of telecommunications and computers, AT & T is becoming embedded in a thicket of inter-corporate relationship. So are biotechnology firms like Genentech and Chiron because the biotechnology research paradigm yields products and processes competitive with pharmaceuticals and chemicals, to name just two examples [Pisano, Shan and Teece (1988), Orsenigo (1989)]. If learning were slower, then there is a good possibility that firms could diversify internally without the assistance of interfirm agreements. Unlike conglomerates, network firms are likely to exhibit survival properties so long as particular technologies continue to collide.

Hypothesis Six: Hollow Corporations

With converging path dependencies, opportunities may arise for entrepreneurs to use contractual mechanisms to quickly assemble diverse capabilities directed at the development and commercialization of a particular product. We refer to such entities as hollow corporations. Unless capabilities develop to undergird these structures, and unless distinctive organizational routines emerge to 'glue' such organizations together, such organizations will not survive, except where the selection environment is weak. If they do survive, it is likely because they are constantly configuring and reconfiguring their alliances. [25]

These hypotheses are summarized in figs. 2a and 2b which are an obvious simplification of the theory we have tried to outline. Note that we have suggested that coherence is a function of several classes of variables: (a) enterprise learning (which we posit is necessarily local in scope); (b) path dependencies, which we see being shaped both by where the firm is currently situated (its existing competences and supporting complementary assets) and the technological opportunities which lie ahead or to one side or the other; and (c) the selection environment, which is a rough measure of the external and internal competition facing the various products the firm produces. We have collapsed these classes of variables in order to represent the predictions of the theory in a fashion which is easy to follow. Note that fig. 2b (weak selection environment) incorporates the organizational forms of fig. 2a because organizational forms that can exist with tight selection can surely exist with weak selection. Hence, fig. 2b contains only the additional, and in our view ultimately nonsustainable, organizational forms - the conglomerate and the hollow corporation. The conglomerate is not associated with environments in which there is rapid learning and innovation because hierarchical structures are incompatible with quick (decentralized) decision making and

[24] As compared to hollow corporations, network firms are assumed to have a core competence. [25] Miles and Snow (1986) describe what they call 'dynamic networks' and suggest they are a new organizational form created in the 1980s. We concur; but except in limited circumstances such as the fashion industry doubt they are viable.

high power incentives. Clearly, we are assuming weak competition and not its complete absence, because with the complete absence of competition any organizational form would be possible.

Remarks on Organizational Mutations

The conglomerate is not viewed especially favorably in our framework. There remains a class of highly diversified conglomerate firms that warrant further comment. The histories of such firms are invariably marked by a long sequence of mergers. Since, however, similar histories exist for many firms that are not highly diversified, growth by merger is not by itself an anomalous fact. The anomaly is the extreme degree of the diversification itself, as illustrated by firms like Tenneco, ITT and LTV. Such examples certainly appear to challenge an evolutionary interpretation of large firms as being, typically, the results of a historical process that involved the success, replication, modification and extension of the competence embedded in some set of routines.

Two responses to this challenge may be offered. The first is to raise the question of the (theoretically relevant) boundaries of the firm. Vesting the ownership of a diverse range of businesses in a common corporate parent may have few effects of real significance as regards the evolution of productive competence. The individual businesses, or sub-units within them, may be the packages of competence that matter.

The history of the Hertz Corporation provides some suggestive evidence in this regard. Hertz has had quite a number of different owners in its approximately 70 year history. Within recent years it was bought by United

Airlines (later renamed Allegis Corporation) from RCA, then was sold to Park Ridge Corporation, an entity formed by Hertz management and the Ford Motor Company to effect a leveraged buyout of Hertz. [26] Yet during this period Hertz operated under its various corporate owners in much the same fashion that it did when it was a freestanding company (1954-1967).

Under RCA, for example, Hertz paid out 50 percent of net income to RCA as dividends and kept the remainder for reinvestment. RCA's management and board were periodically apprised of Hertz's activities and plans. RCA was concerned with personnel policy issues at Hertz and, in particular, wanted Hertz to fill high-level managerial and staff positions with people from its own divisions. Matters of direct relevance to the car rental business remained Hertz's province throughout. RCA bought Hertz when conglomerate diversification was in favor, and sold it in a retreat to the 'core business' when such diversification fell out of favor. If one asks where the competence in the car rental business resided and what shaped its evolution, the answers seem clear: the competence resided in Hertz. Rivals, other environmental factors, and Hertz's own innovative efforts shaped the evolution of that competence much more strongly than Hertz's corporate parents did.

One major school of thought on the conglomerate corporation holds that its primary economic function is to serve as a 'miniature capital market' [Williamson (1975, 1985)], reallocating funds from lines of activity that generate high current cash flow to those whose future prospects and current investment needs outstrip currently available resources. Under this interpretation, relations between corporate headquarters and the individual business units within the conglomerate structure largely involve financial flows and financial monitoring. To the extent that the ownership rclation facilitates monitoring and appraisal of future prospects relative to what is possible for the external capital market, the conglomerate form may have a secure niche in the long-term institutional ecology of capitalism. [27]

This appraisal is not obviously in conflict with the view that it is individual business units that are the important repositories of technicalproductive competence, as illustrated by the example of Hertz. No judgment is implied as to the quality of the financial management of conglomerates

in any absolute sense; there is only a claim that they may be sufficiently effective relative to the available institutional alternatives to account for their existence. Although the selection process would function differently than if there were no conglomerates, no judgment is implied as to how much different it would be. On this view, the clearest implication of conglomerates for evolutionary analysis is that they impede investor research: instead of individual financial statements for each of a number of independent businesses, we encounter a single set of financial statements for the conglomerate. This certainly tends to hamper efforts to assess the performance of the individual businesses.

The alternative response to the analytical challenge represented by large conglomerates involves a more negative appraisal of the conglomerate form and its future prospects. Biologists have employed the term 'hopeful monsters' to refer to new types arising as a result of a fundamental 'systematic mutation', among which a few occasionally have characteristics that are compatible with survival and result in the founding of a new species. Nothing in the laws of biological evolution precludes the appearance of such types; on the other hand, one does not expect to see the fossil record strewn with examples because natural selection promptly eliminates all but a tiny fraction of them. On the average, there is not much hope per monster. The same may be true of large conglomerate corporations.

Techniques of financial control and monitoring are, on this view, good enough to lend a temporary surface plausibility to the conglomerate form but not good enough to assure its long-term viability in competition with more specialized organizations. In contrast to its biological analogues, the typical conglomerate monster does not expire soon after its creation. Its odd morphology is nonfatal precisely because it is so remote from being an integrated whole; it involves the binding together of entities that are viable, or approximately so, on their own. Furthermore, the power of corporate headquarters in the financial realm carries with it a significant amount of discretion regarding the reporting of results (not to speak of the discretion to make headquarters, at least, look prosperous). A long-term unfavorable trend in performance may be disguised for an extended period. The fundamental question is whether headquarters is paying its way, and the suggestion here is that it is not. [28]

To sort out the evidence regarding the performance of the conglomerate form, and the broader issues regarding diversification generally, is a larger undertaking than can be attempted herc. [29] The 'hopeful monsters' interpretation is most congenial to evolutionary economic theory. A demonstration of its validity would, on the one hand, tend to support our view of the firm as a package of technical and organizational competences. When the contents of the package lack an organizational competence, the twine of financial control will rarely suffice to bind together a successful enterprise.

We likewise view the hollow corporation as an organizational mutation. If a firm is in fact merely a nexus of contracts, there is little opportunity for the various pieces to engage in mutual learning. This is because the contracting units are likely to have multiple contractual partners, further complicating transaction specific learning. A good example of a hollow corporation was

[26] An interesting contrast in the durability of transacting patterns is provided by the case of O.J. Simpson who has been doing Hertz commercials for over a decade. [27] In periods when tax preferences for capital gains or other institutional factors impede the functioning of the external capital markets, conglomerates may be expected to play a larger role than at times when such biases are absent.

[28] In the terms used in Williamson's (1985, pp. 286-290) discussion of conglomerates, the suggestion here is that the extension of M-form managerial techniques to the control of higher diverse activities may typically be unsuccessful - though an occasional monster may do well, at least for a while. [29] Evidence available as of the mid-1980s in the United States suggests that conglomerates are divesting significant assets. The average number of business centered in conglomerates was declining, while business relatedness in the conglomerate was increasing [Williams, Paez and Sanders (1986)].

Worlds of Wonder, an electronic toy company founded in Hayward, California in 1984. The company employed a strategy of contracting for almost everything - manufacturing, new product development, distribution, and marketing. The company grew to over $100 million in sales in record time, and then disappeared. There were a variety of reasons for the company's failure, not least of which was its absence of follow-along products. But the company's extreme dependency on contracts helps explain both its rapid growth and its rapid demise. It may have briefly possessed organizational competence, but it had no technical-productive competence of its own, and the former was not enough to sustain it in the business in which it ever so briefly competed.

Conclusion

We have shown that as firms grow more diverse, overall coherence does not change much in the United States. However, there are large differences among firms with respect to the degree of their coherence. We hypothesize that some of these differences can be explained as a complex interaction between three classes of factors: (1) enterprise learning, which we suggest is a 'local' phenomena; (2) evolutionary paths, which we see being shaped both by where firms have been (in terms of their scope of their past activities), where they are now (their current competences and complementary assets), and the opportunities which lie ahead; and (3) the strength of competition and the mechanism firms have to shield themselves from it, whether in the product market or the capital market. We have shown how these factors can be expected to mold the scope of a firm's activities. Given tight selection environments, we conclude that coherent diversifiers, who expand by adding related activities, are likely to be the most common corporate mode. Indeed, we find for the U.S. economy as a whole that diversification is of this kind.

We suggest that we have provided a framework to explain central tendencies in the growth of the business enterprise. We have assembled what we believe are provocative hypotheses, as well as some suggestive evidence. Our hope is that others will be both intrigued and challenged by our findings and our theorizing, and that as a consequence we will soon have a deeper understanding of one of the great theoretical conundrums in industrial organization - the coherence of the multiproduct business firm.

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